

Digital Communication System



Unit 4: Wireless Enterprise Network

- 4.1 Bluetooth technology
- 4.2 RFID technology
- 4.3 Mobile IP
- 4.4 Infrared communication technology (covered in Unit2)
- 4.5 Wireless sensor networks
- 4.6 WIFI, WIMAX Technology

Bluetooth

- **Bluetooth** - Bluetooth is universal for short-range wireless voice and data communication. It is a Wireless Personal Area Network (WPAN) technology and is used for exchanging data over smaller distances.



Symbol of Bluetooth



An example of a Bluetooth device

Bluetooth



Your smart phone transmits around a million 1s and 0s to the head phones every second via Bluetooth

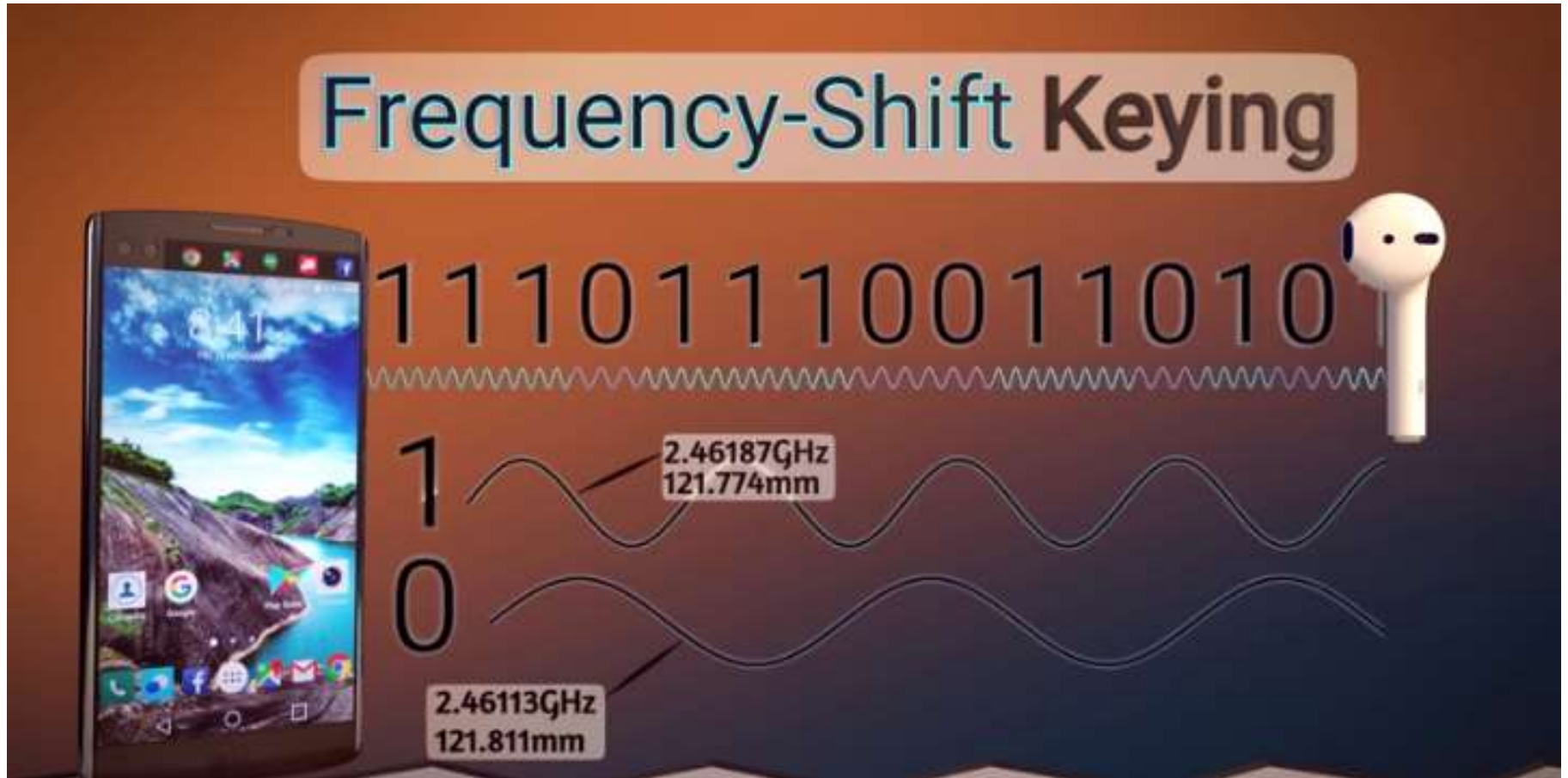
Bluetooth



Bluetooth microchip filters noise, detects error, does frequency hopping, and converts data into packets

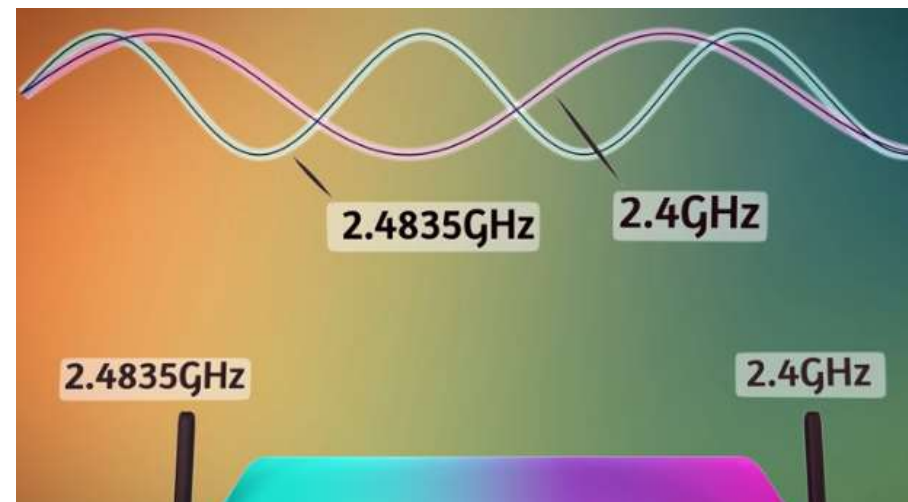
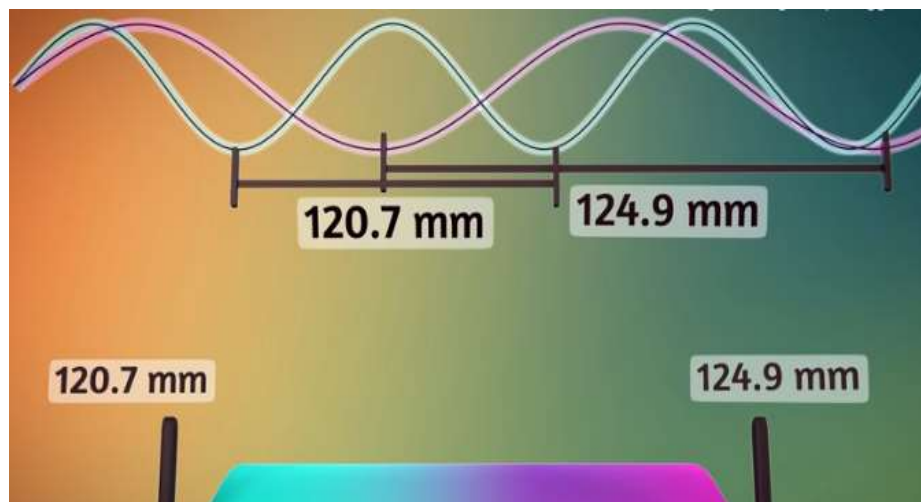
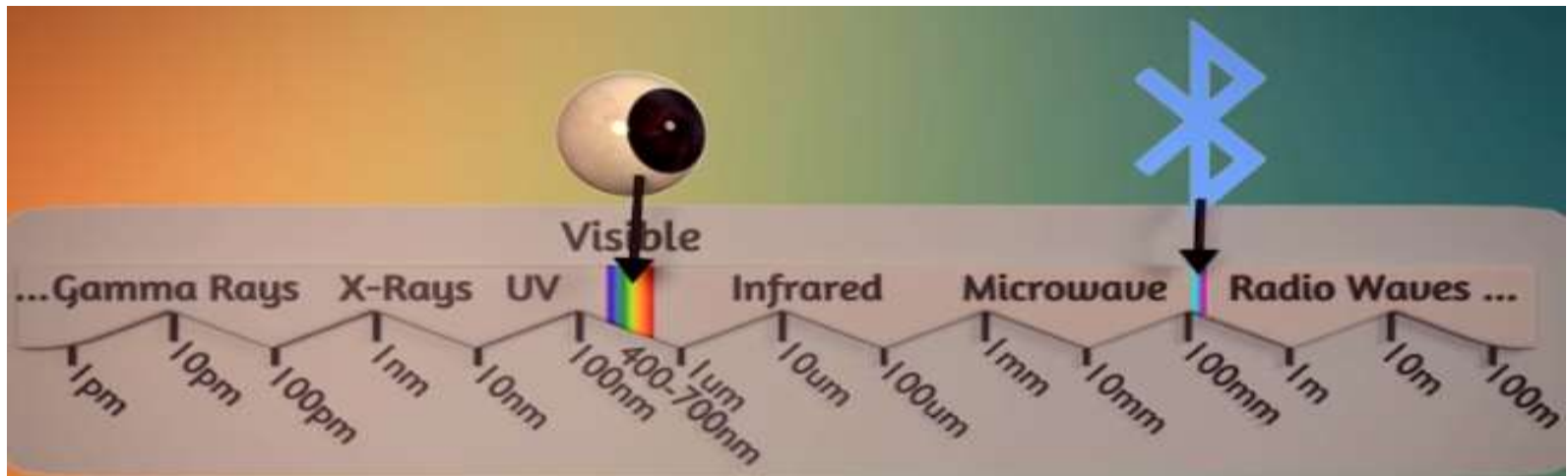
Bluetooth

Frequency-Shift Keying



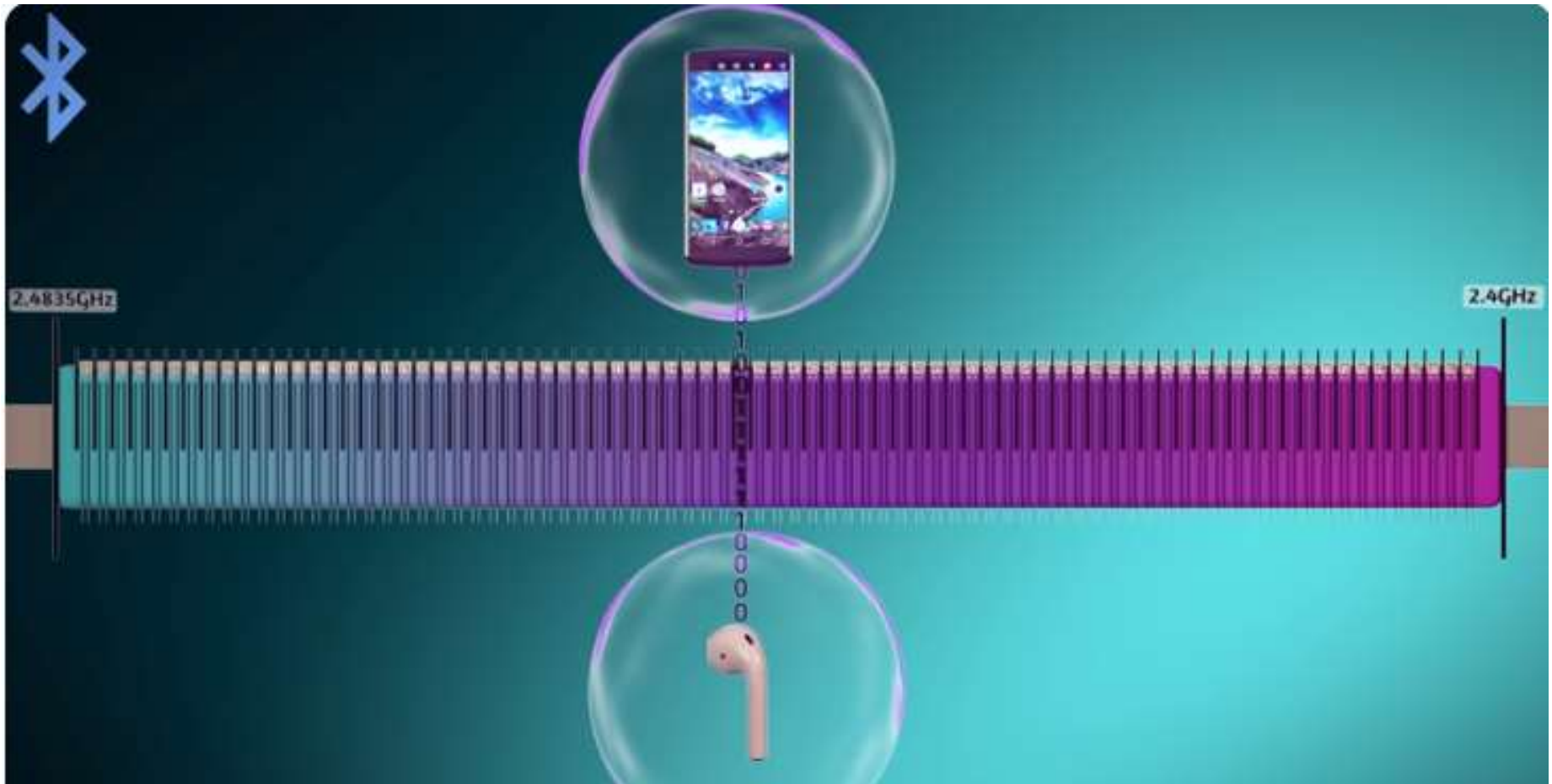
Modulation Technique

Bluetooth



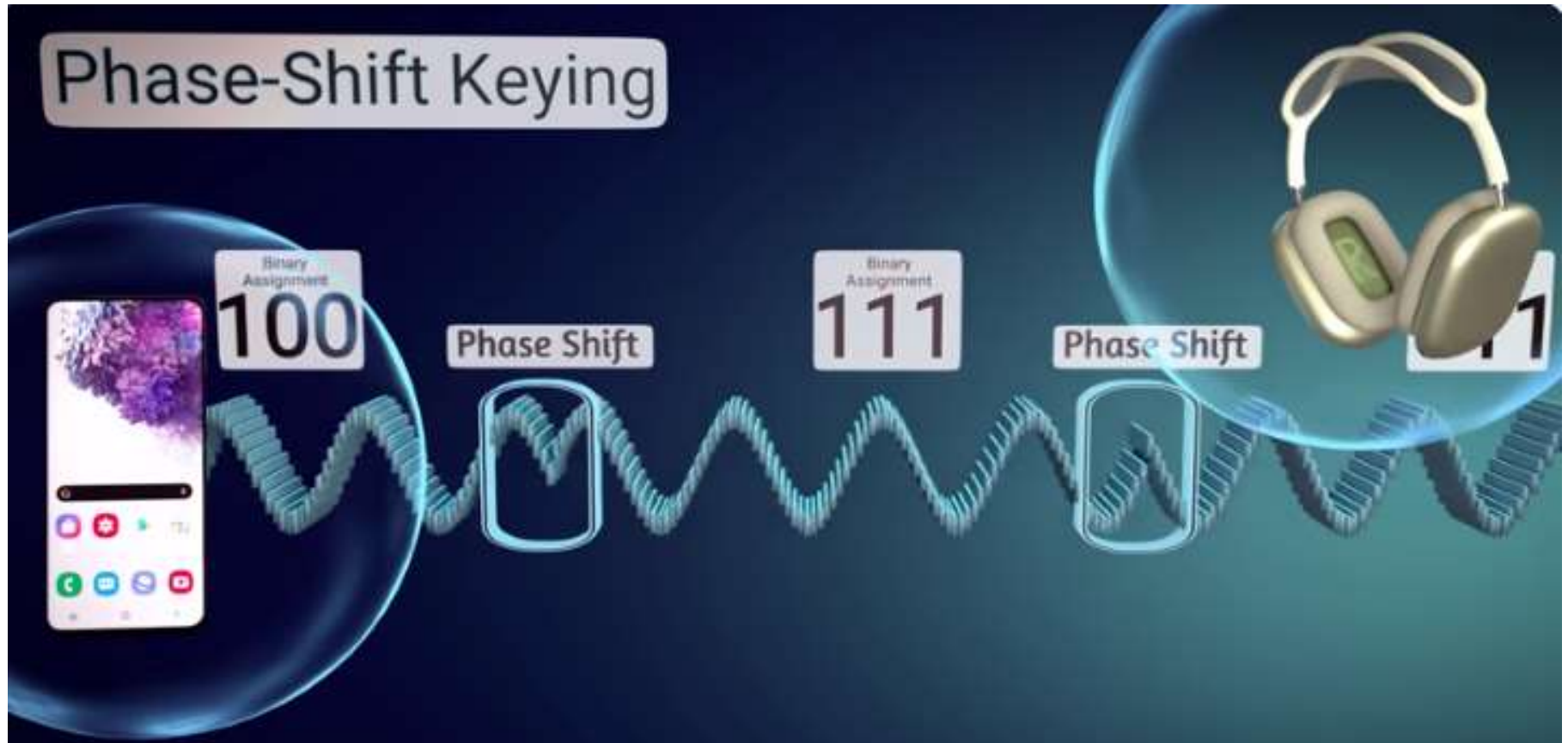
Frequency range 2.4GHz to 2.4835GHz

Bluetooth



Frequency bandwidth is divided into number of channels and each pair of sender and receiver works on a specific frequency channel

Bluetooth



Modulation Technique

Bluetooth

- This technology was invented by Ericson in 1994.
- It operates in the unlicensed, industrial, scientific, and medical frequency (ISM) band from 2.4 GHz to 2.485 GHz.
- Maximum devices that can be connected at the same time are 7.
- Bluetooth ranges up to 10 meters.
- The spreading technique that it uses is FHSS (Frequency-hopping spread spectrum) for handling interference and interception.

Bluetooth Applications

- It can be used in laptops, and in wireless PCs, printers.
- It can be used in wireless headsets, wireless PANs, and LANs.
- It can connect a digital camera wirelessly to a mobile phone.
- It can transfer data in terms of videos, songs, photographs, or files from one cell phone to another cell phone or computer.
- It is used in the sectors of Medical health care, sports and fitness, Military.

Bluetooth Advantages

- It is a low-cost and easy-to-use device.
- It can also penetrate through walls.
- It creates an Ad-hoc connection immediately without any wires.
- It is used for voice and data transfer.

Bluetooth Disadvantages

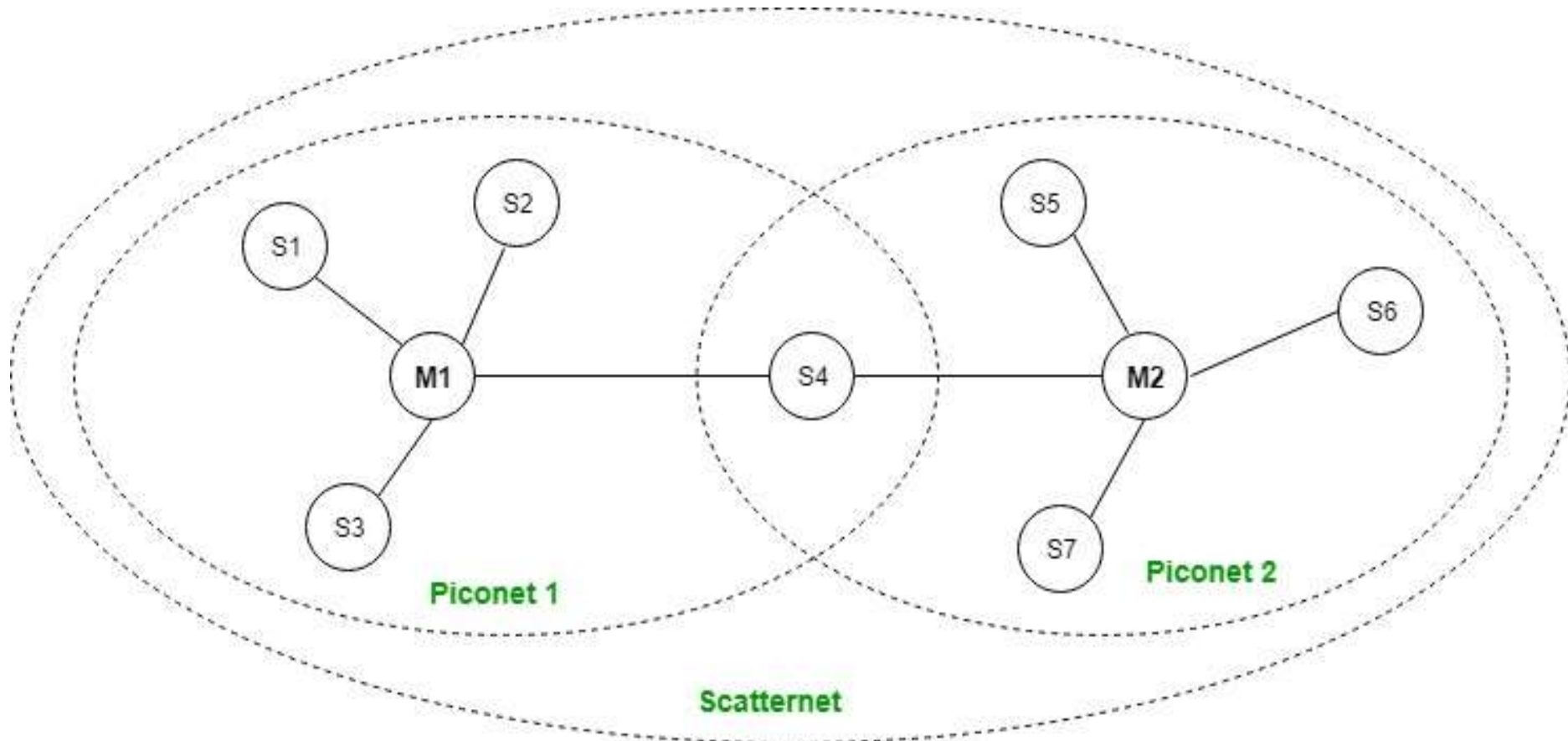
- It can be hacked and hence, less secure.
- It has a slow data transfer rate: of 3 Mbps.
- It has a small range: 10 meters.
- Bluetooth communication does not support routing.

Types of Bluetooth

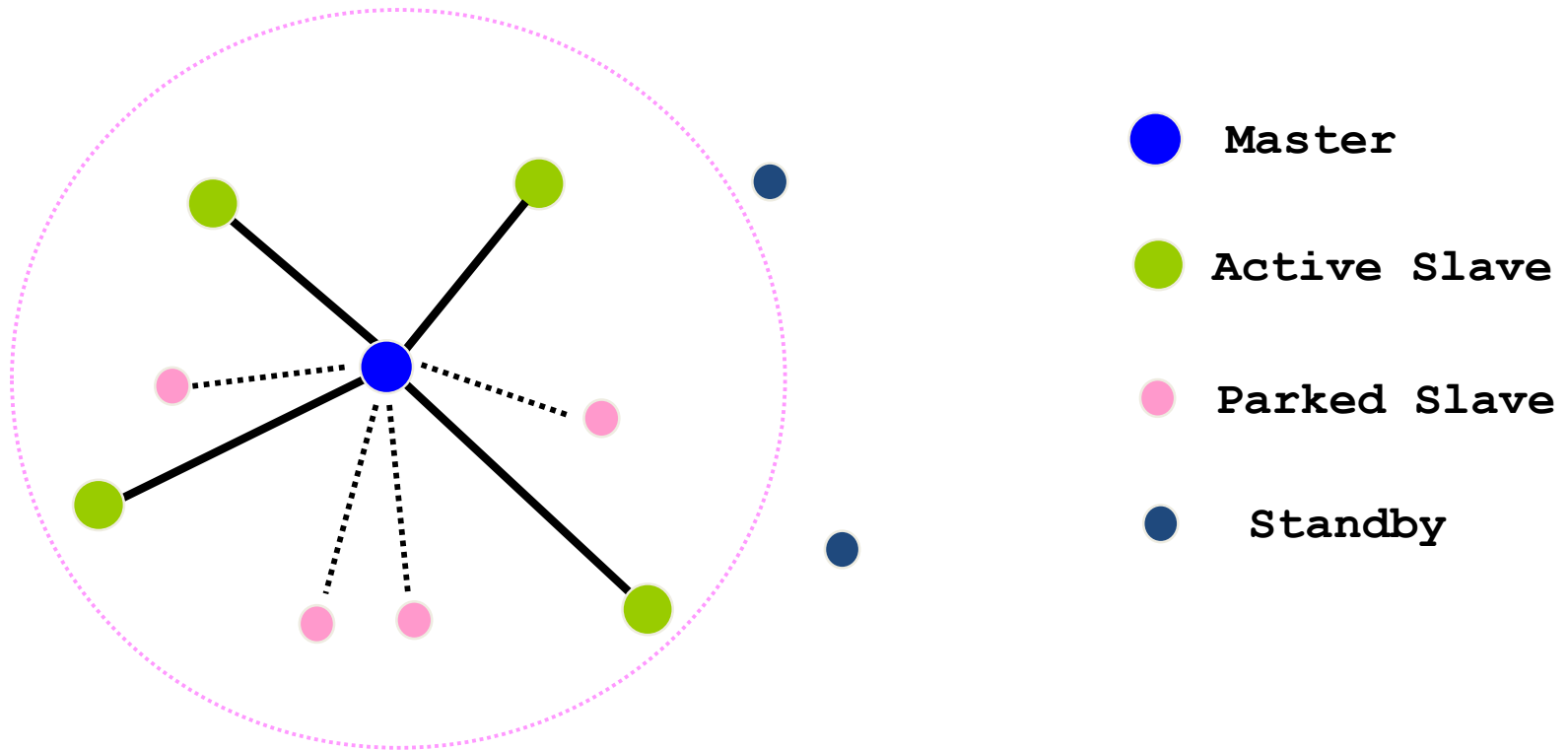
- **Stereo Headset:** To listen to music in car or in music players at home.
- **Webcam:** One can link the camera with the help of Bluetooth with their laptop or phone.
- **Bluetooth-equipped Printer:** The printer can be used when connected via Bluetooth with mobile phone or laptop.
- **Bluetooth Global Positioning System (GPS):** To use GPS in cars, one can connect their phone with car system via Bluetooth to fetch the directions of the address.

Bluetooth Architecture

A Bluetooth network is called a **piconet** and a collection of interconnected piconets is called **scatternet**.



Bluetooth Architecture



- All devices in piconet hop together.
- Master's ID and master's clock determines frequency hopping.

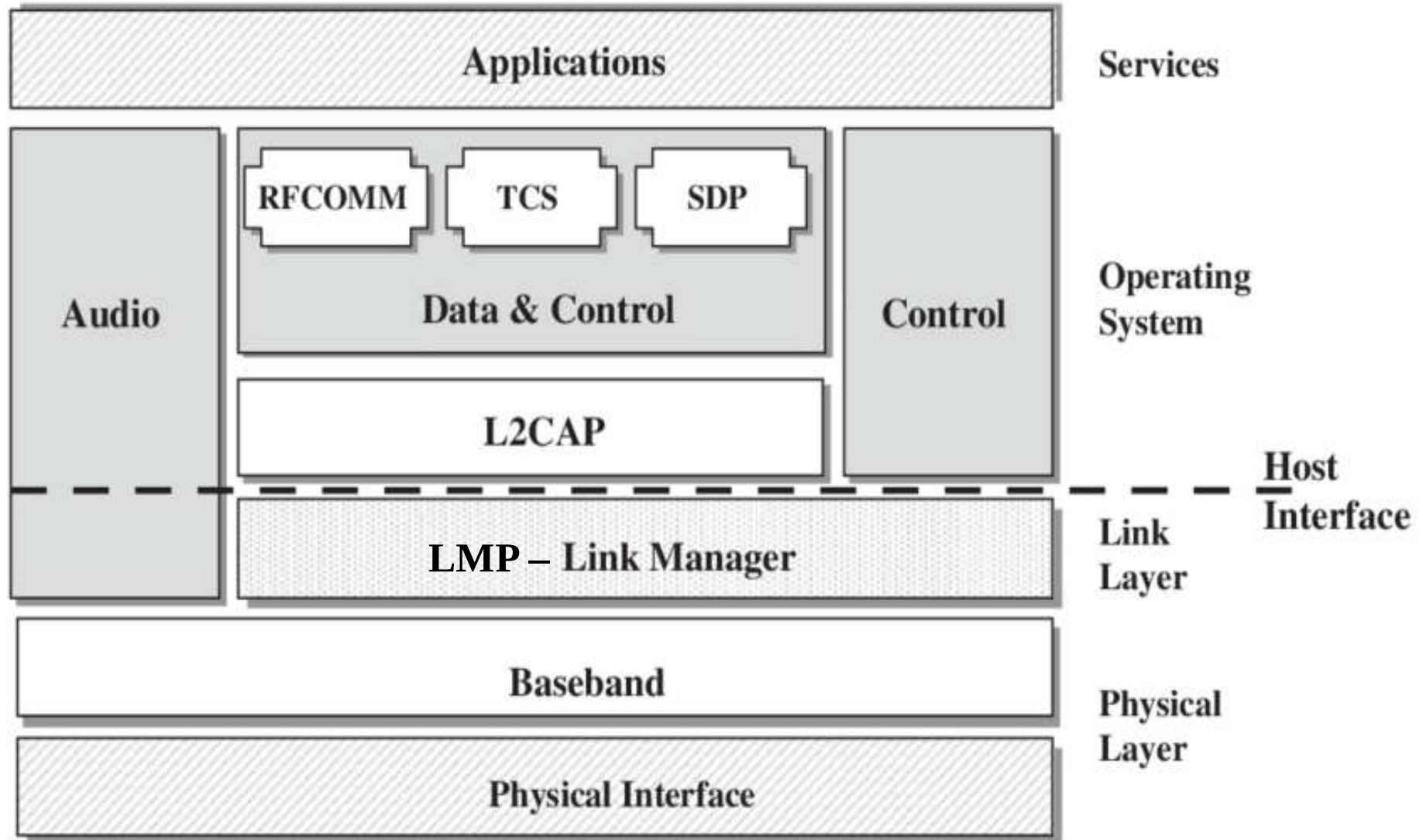
Bluetooth Architecture

- **Piconet** - Piconet is a type of Bluetooth network that contains **one primary node** called the **master node** and **seven active secondary nodes** called **slave nodes**. Thus, we can say that there is a total of 8 active nodes which are present at a distance of 10 meters.
- The communication is only between the master and slave. It can be one to one or one to many.
- It also has **255 parked nodes**, these are secondary nodes and cannot take participation in communication unless it gets converted to the active state.

Bluetooth Architecture

- **Scatternet** - It is formed by using **various piconets**.
- A slave that is present in one piconet can act as master or we can say primary in another piconet. This kind of node can receive a message from a master in one piconet and deliver the message to its slave in the other piconet where it is acting as a master. This type of node is referred to as a **bridge node**.
- A station cannot be mastered in two piconets.

Bluetooth Protocol Stack



Bluetooth Protocol Stack

- **Physical Interface/ Physical Radio** – Defines bluetooth transceiver devices and characteristics .
- This is a physical layer equivalent protocol that lays down the physical structure and specifications for transmission of radio waves. It defines air interface, frequency bands, frequency hopping specifications and modulation techniques.
- **Baseband** – Link establishment and power control.
- This protocol takes the services of radio protocol. It defines the addressing scheme, packet frame format, timing, and power control algorithms.

Bluetooth Protocol Stack

- **Link Manager (LMP)** – Responsible for link setup .
- It performs the management of the already established links which includes authentication and encryption processes. It is responsible for creating the links, monitoring their health, and terminating them gracefully upon command or failure.
- **Logical Link Control and Adaption Protocol layer (L2CAP)**
 - It is also known as the **heart of the Bluetooth protocol stack**. It allows the communication between upper and lower layers of the Bluetooth protocol stack. It packages the data packets received from upper layers into the form expected by lower layers. It also performs segmentation and multiplexing.

Bluetooth Protocol Stack

Service Discovery Protocol (SDP) layer - It is short for Service Discovery Protocol. It allows discovering the services available on another Bluetooth-enabled device.

RF comm layer - It is a cabal replacement protocol (makes bluetooth wireless). It is short for Radio Frontend Component. It provides a serial interface with WAP and OBEX. It also provides emulation of serial ports over the logical link control and adaption protocol(L2CAP). The protocol is based on the ETSI standard TS 07.10.

Bluetooth Protocol Stack

- **Object Exchange (OBEX)** - It is a communication protocol to exchange objects between 2 devices.
- **Wireless Access Protocol (WAP)** - It is used for internet access.
- **Telephony Control Protocol(TCS)** - It provides telephony service. The basic function of this layer is call control (setup & release) and group management for the gateway serving multiple devices.
- **Application layer** - It enables the user to interact with the application.

RFID

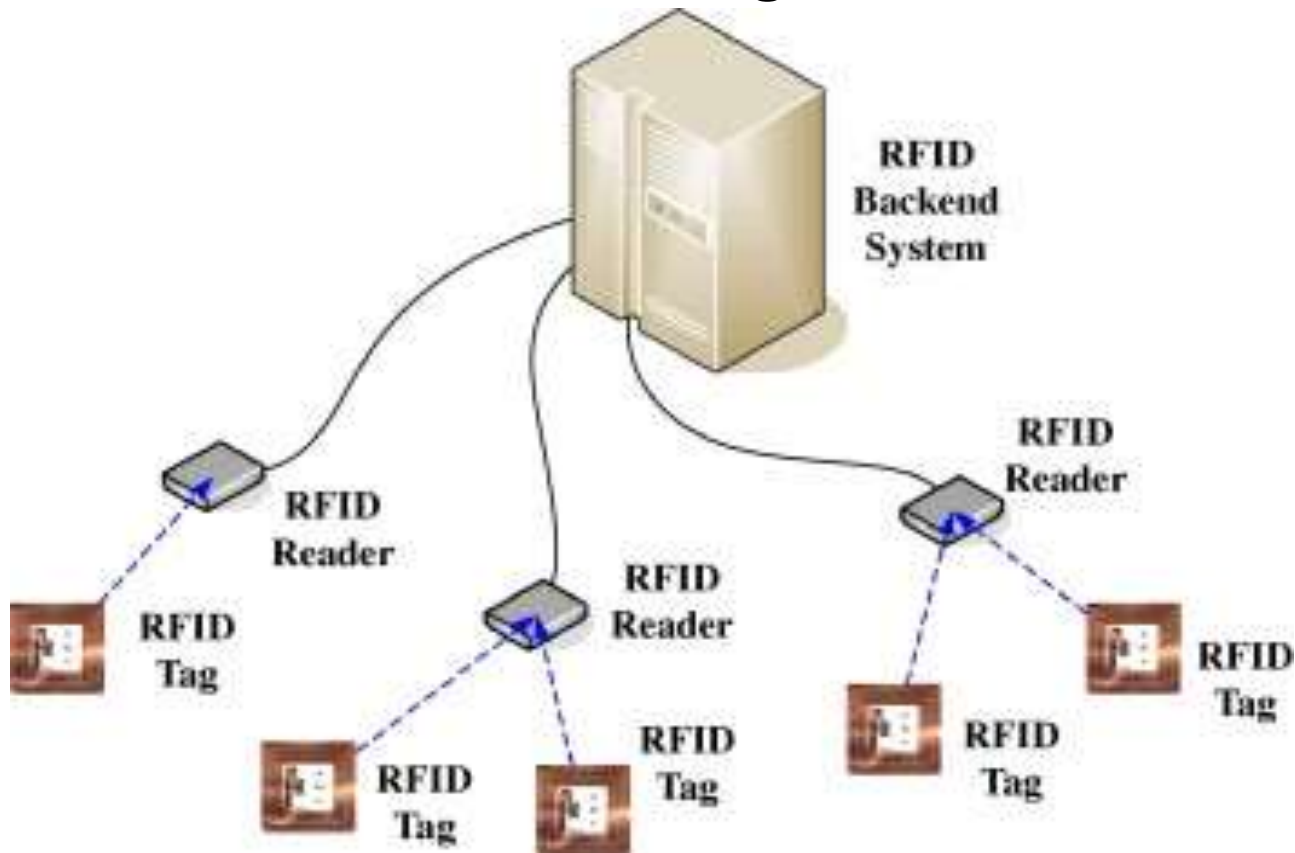
- **Radio Frequency Identification** – RFID is a wireless technology that uses radio waves to passively identify a tagged object, person or animal.
- Radio Frequency Identification (RFID) refers to **a wireless system comprised of two components: tags and readers.** The reader is a device that has one or more antennas that emit radio waves and receive signals back from the RFID tag.

RFID

- **Radio Frequency Identification** – RFID is a wireless technology that uses radio waves to passively identify a tagged object, person or animal.
- **Is fast** and does not require physical sight or contact between reader/scanner and the tagged item.
- **It is cost effective** - Performs the operation using low cost components.

RFID

- Radio Frequency Identification (RFID) refers to **a wireless system comprised of two components: tags and readers.** The reader is a device that has one or more antennas that emit radio waves and receive signals back from the RFID tag.



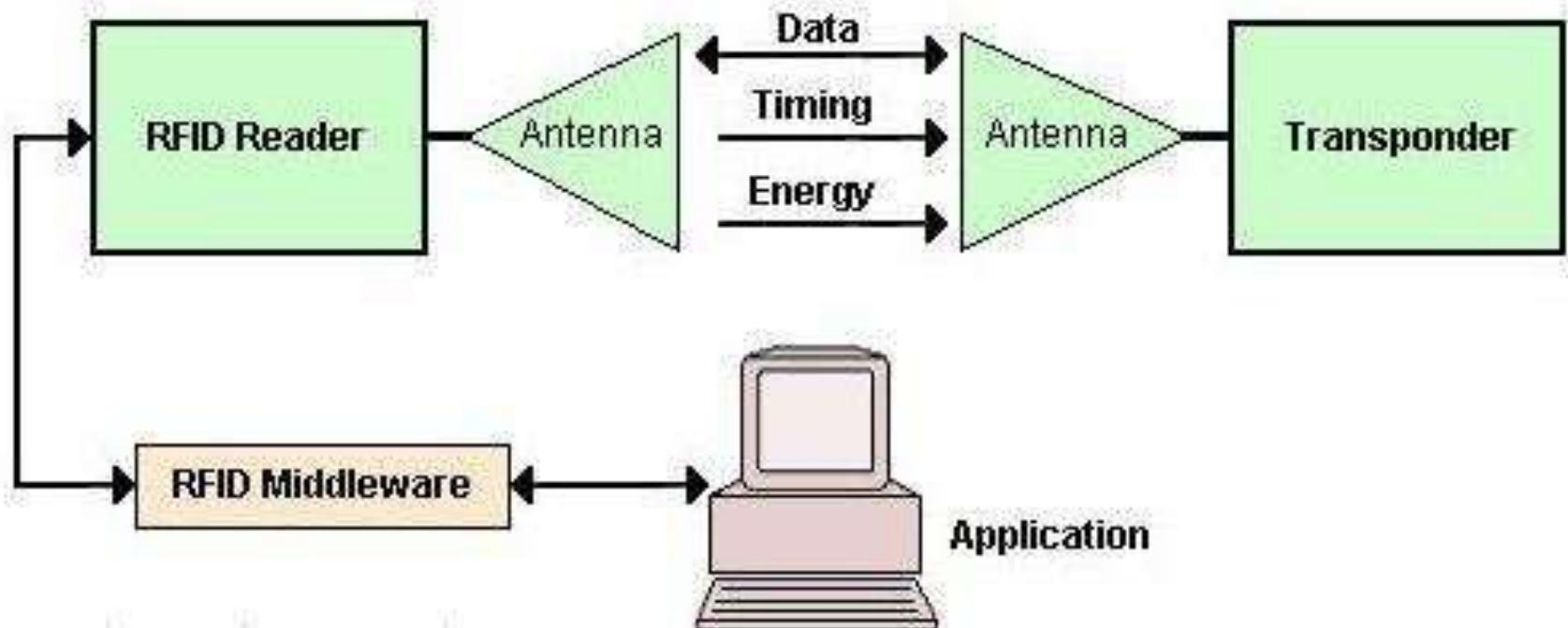
RFID Working

Every RFID system consists of three components:

a scanning antenna

a transceiver (RFID Reader)

a transponder (RFID Tag)



RFID Working

A Scanning Antenna - It scans RFID

RFID Reader

- When the scanning antenna and transceiver are combined, they are referred to as an RFID reader or interrogator.
- There are two types of RFID readers -- fixed readers and mobile readers.
- It uses radio waves to transmit signals that activate the tag. Once activated, the tag sends a wave back to the antenna, where it is translated into data.

RFID Working

RFID Transponder

- The transponder is in the **RFID tag** itself.
- The read range for RFID tags varies based on factors including the type of tag, type of reader, RFID frequency and interference in the surrounding environment or from other RFID tags and readers.
- Tags that have a stronger power source also have a longer read range.

RFID Working

- **RFID Tags are of two types:**

- **Passive Tags – Do not require Battery**

- Do not require power – Draws from Interrogator Field
- Lower storage capacities (few bits to 1 KB)
- Shorter read ranges (4 inches to 15 feet)
- Usually Write-Once-Read-Many/Read-Only tags
- Cost around 25 cents to few dollars

- **Active Tags - Requires Battery**

- Battery powered
- Higher storage capacities (512 KB)
- Longer read range (300 feet)
- Typically can be re-written by RF Interrogators
- Cost around 50 to 250 dollars

RFID Applications

Manufacturing and Processing

- Inventory and production process monitoring

- Warehouse order fulfillment

Supply Chain Management

- Inventory tracking systems

- Logistics management

Retail

- Inventory control and customer insight

- Auto checkout with reverse logistics

Security

- Access control

- Counterfeiting and Theft control/prevention

Location Tracking

- Traffic movement control and parking management

- Wildlife/Livestock monitoring and tracking

Real World examples of RFID Applications

- Fast Tag for vehicles
- Tags on clothes in Malls
- Tags on Animals
- Tags for security checks to control entry exit
- Baggage tracking in Aviation
- Healthcare

Wi – Fi (Wireless Fidelity)

- **Wi-Fi** is a wireless technology used to connect computers, tablets, smart phones and other devices to the internet. Wi-Fi is the radio signal sent from a wireless router to a nearby device, which translates the signal into data you can see and use.



Wi – Fi (Wireless Fidelity)

- Because they communicate with each other over airwaves, your devices and personal information can become vulnerable to hackers, cyber-attacks and other threats.
- This is especially true when you connect to a public Wi-Fi network at places like a coffee shop or airport.
- When possible, it's best to use Wi-Fi that is **password-protected or a personal hotspot.**

Wi – Fi (Wireless Fidelity)

- Because they communicate with each other over airwaves, your devices and personal information can become vulnerable to hackers, cyber-attacks and other threats.
- This is especially true when you connect to a public Wi-Fi network at places like a coffee shop or airport.
- When possible, it's best to use Wi-Fi that is **password-protected or a personal hotspot.**

Wi – Fi (Wireless Fidelity)

- **Wi-Fi** is also known as **WLAN** (wireless local area network). The communication standard is **IEEE 802.11**.
- WiFi technology allows local area networks to operate without cable and wiring.
- It is making a popular choice for home and business networks.
- A computer's wireless adaptor transfers the data into a radio signal and transfers the data into an antenna for users.

Wi - Fi Working

- There 3 main components of Wi Fi are:
- Base Station
- Access Point (Routers)
- Accessing Devices (Receivers)



Wi – Fi Working

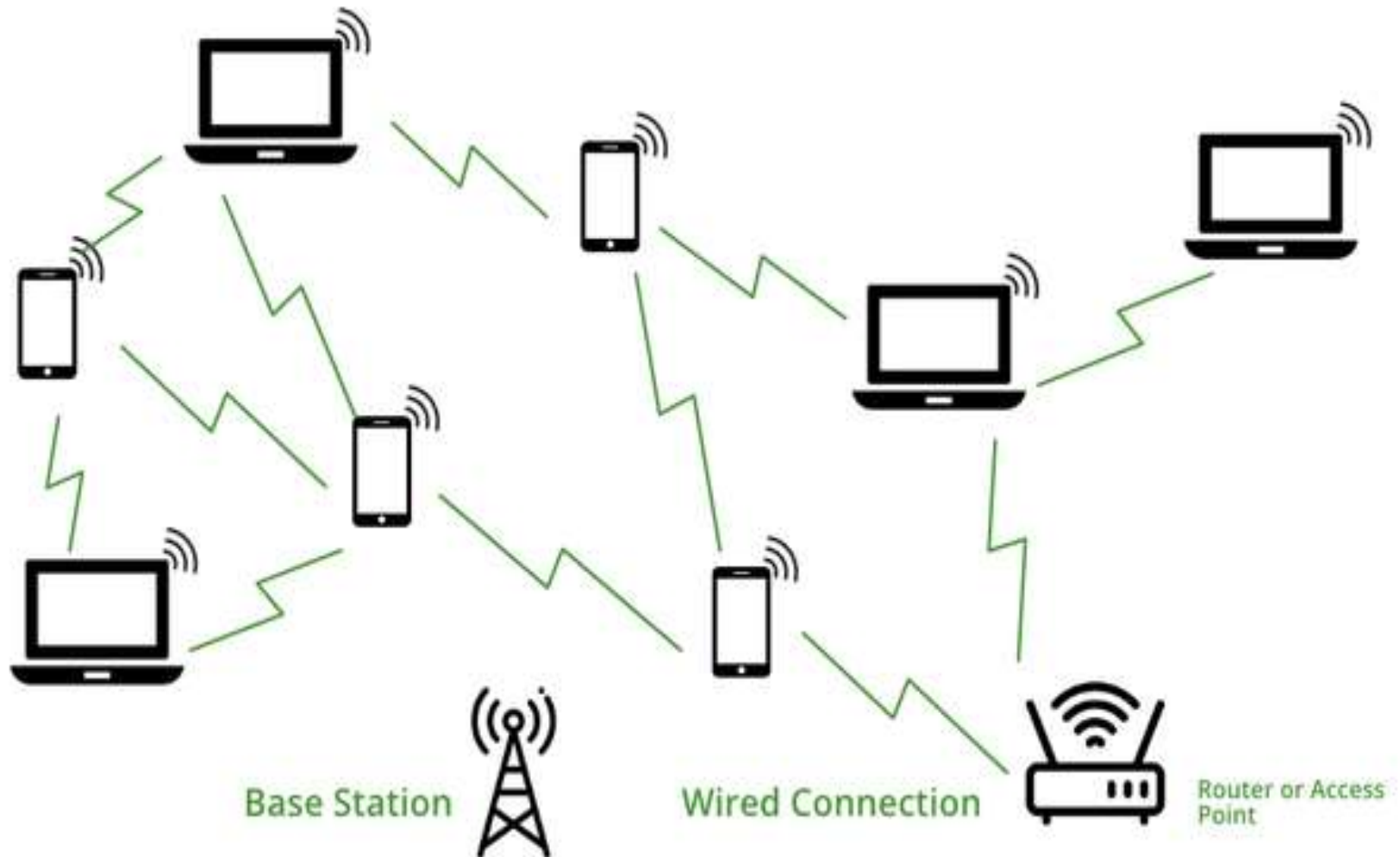
- **Main Components of the Wi Fi System:**

Base station network or an Ethernet(802.3) connection: It is the main host network from where the network connection is provided to the router.

Access point or router: it is a bridge between a wired network and a wireless network. It accepts a wired Ethernet connection and converts the wired connection to a wireless connection and spreads the connection as a radio wave.

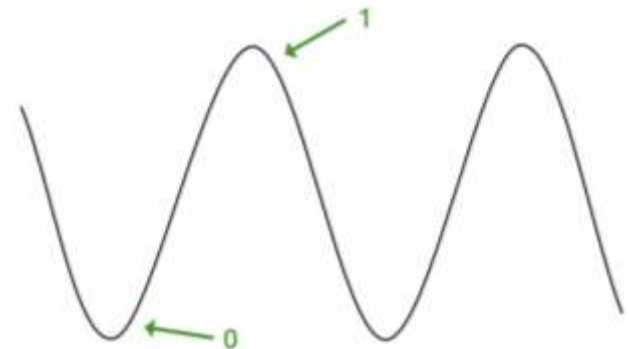
Accessing devices: It is our mobile, computer, etc from where we use the Wi-Fi and surfing internet.

Wi - Fi Working



Wi – Fi Working

- A single access point can support up to 30 users and can function within a range of 100 – 150 feet indoors and up to 300 feet outdoors.
- All the electronics devices read data in binary form, also router or our devices, here routers provide radio waves and those waves are received by our devices and read the waves in binary form. We all know how a wave looks like, the upper pick of the wave is known as 1 and the lower pick of the wave is known as 0 in binary.



Wi – Fi Name

SSID (Service Set Identifier): It is a 32 character name that identifies the Wi-Fi network and differentiates one Wi-Fi from another Wi-Fi. All the devices are attempting to connect a particular SSID. Simply, **SSID is the name of the wireless network.**

Wi – Fi Security

WPA-PSK (Wi-Fi Protected Access- Pre-Shared Key): It is a program developed by the Wi-Fi Alliance Authority to secure wireless networks with the use of Pre-Shared Key(PSK) authentication. **WPA (Wifi Protected Access)** has 3 types, such as WPA, WPA2, WPA3. It is a way of encrypting the Wi-Fi signal to protect from unwanted users.

- Wi-Fi **uses Ad-Hoc networks** to transmit. It is a point-to-point network without any interface.

Wi – Fi Security

WiFi Monitoring by Security agencies

These WiFi networks can be monitored by Dept of Telecom and other security agencies at any time, such an arrangement is called **Lawful Interception (LI)**.

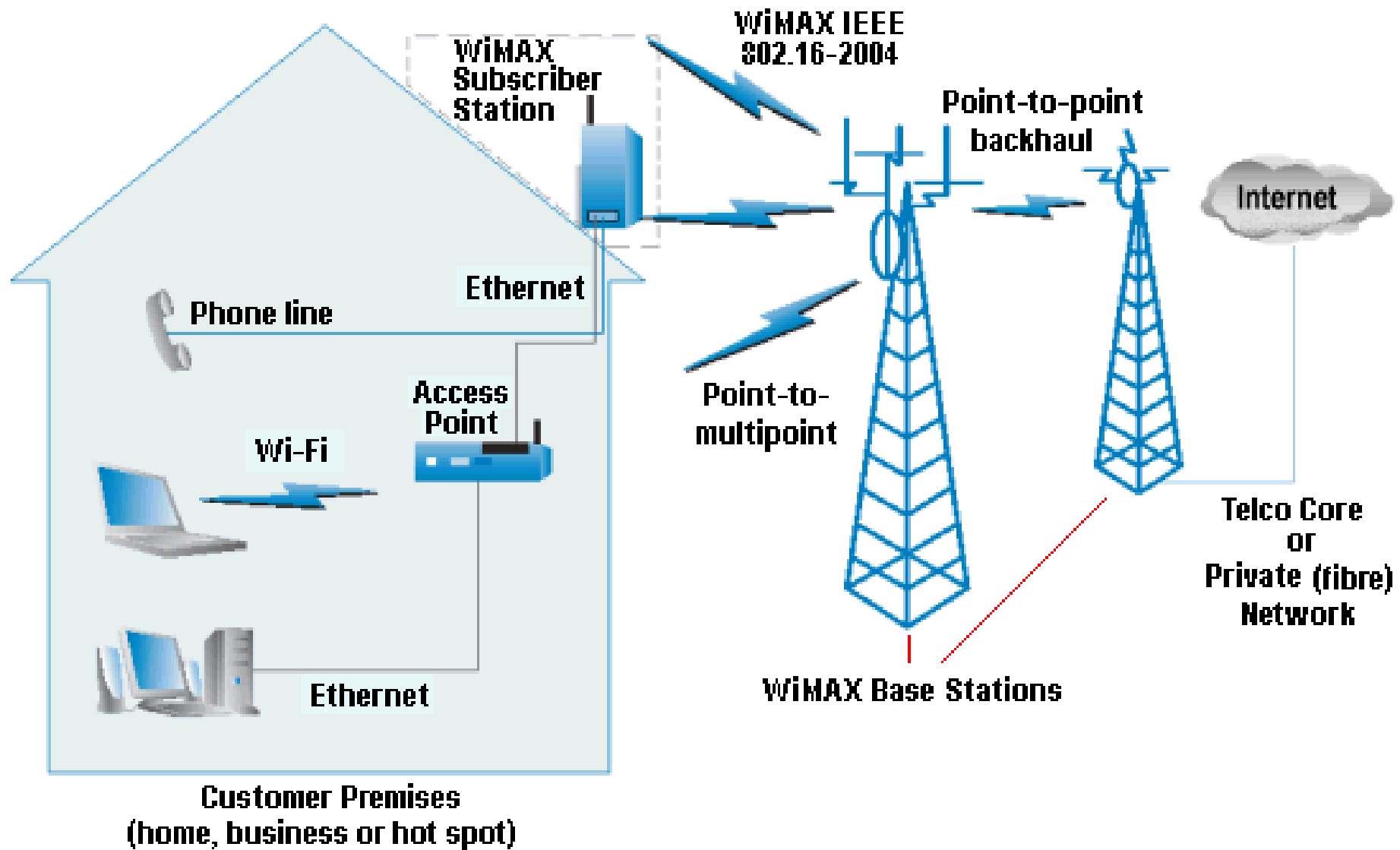
Wi – Fi Advantages

- It is a flexible network connection, no wiring complexities.
- Can be accessed from anywhere in the Wi-Fi range.
- It does not require regulatory approval for individual users.
- It is scalable, can be expanded by using Wi-Fi Extenders.
- It can be set up in an easy and fast way. Just need to configure the SSID and Password.
- Security is high in today's Wi-Fi network.
- It is also lower in cost.

Wi – Fi Disadvantages

- Power consumption is high while using Wi-Fi in any device which has a battery, such as mobile, laptops, etc.
- Speed is slower than a direct cable connection.
- It's radiation can harm humans.
- Wi-Fi signals may be affected by climatic conditions like thunderstorms.
- To use Wi-Fi we need a router, which needs a power source, so at the time of power cut, we cannot access the Internet.

Wi - Max



Wi – Max

- WiMax stands for **Worldwide Inter-operability for Microwave Access**.
- This technology is based on **IEEE 802.16**.
- It is used to provide higher data rates with increased coverage.
- It is based on **MAN (Metropolitan Area Network)** technology. Its range is upto **50 Km**. It may provide speed upto **70 Mbps** and it can operate in Non-Line-of-Sight.
- **This technology is fast, convenient and cost effective.**

Advantages of Wi – Max

Wide Coverage Area (50 km): It is suitable for providing broadband access in rural and underserved areas.

High Data Rates (up to 75 Mbps) : which is higher than many other wireless technologies.

Scalability: WiMAX can be easily scaled to support a large number of users and devices.

Advantages of Wi – Max

- **Interoperability:** WiMAX is based on an international standard, which allows for interoperability between different vendors' equipment.
- **Cost-effective:** WiMAX is a cost-effective solution for providing broadband access in areas where it is not economically feasible to deploy wired infrastructure.

Disadvantages of Wi – Max

- **Interference:** WiMAX operates in the same frequency range as other wireless technologies, which can lead to interference.
- **Security Concerns:** WiMAX uses a shared spectrum, which can make it vulnerable to security threats such as eavesdropping and jamming.
- **Limited penetration:** WiMAX signals may have trouble penetrating through walls, buildings and other obstacles.

Wi-Fi

vs.

Wi – Max

S. No.	Parameters	WiFi	WiMax
1.	Defined	<u>WiFi</u> is defined under IEEE 802.11x standards where x stands for various WiFi versions.	<u>WiMAX</u> is defined under IEEE 802.16y standards where y stands for various WiMAX versions.
2.	Type of Network	WiFi is for LAN (Local Area Network) applications.	WiMAX is for <u>MAN</u> (Metropolitan Area Network) applications.
3.	Guarantee	WiFi does not guarantee any Quality of Service (<u>QoS</u>).	WiMAX guarantees Quality of Service (QoS).

Wi-Fi

vs.

Wi – Max

S. No.	Parameters	WiFi	WiMax
4.	Distance	The WiFi network range is around 100 meters.	WiMAX network can reach about 50-90 km.
5.	Connection oriented	WiFi uses the conflict-based <u>CSMA/CA</u> protocol which is not connection-oriented.	WiMAX networks operate on a connection-oriented <u>MAC</u> .
6.	Range	WiFi is short-range technology.	WiMAX is long-range technology.

Wi-Fi

vs.

Wi – Max

S. No.	Parameters	WiFi	WiMax
7.	Transmit	WiFi connection can transmit up to 54 Mbps.	WiMAX connection can transmit up to 70 Mbps.
8.	Signal	WiFi access points transmit signals to the receiving device.	A signal will be sent from the WiMAX base station or tower station to the WiMAX receiver.
9.	Cost	It is inexpensive compared to WiMAX technology.	It is expensive because of high installation cost.
10.	Usage	WiFi is utilized for domestic use and short-distance connections.	WiMAX supports a broader interoperable network,

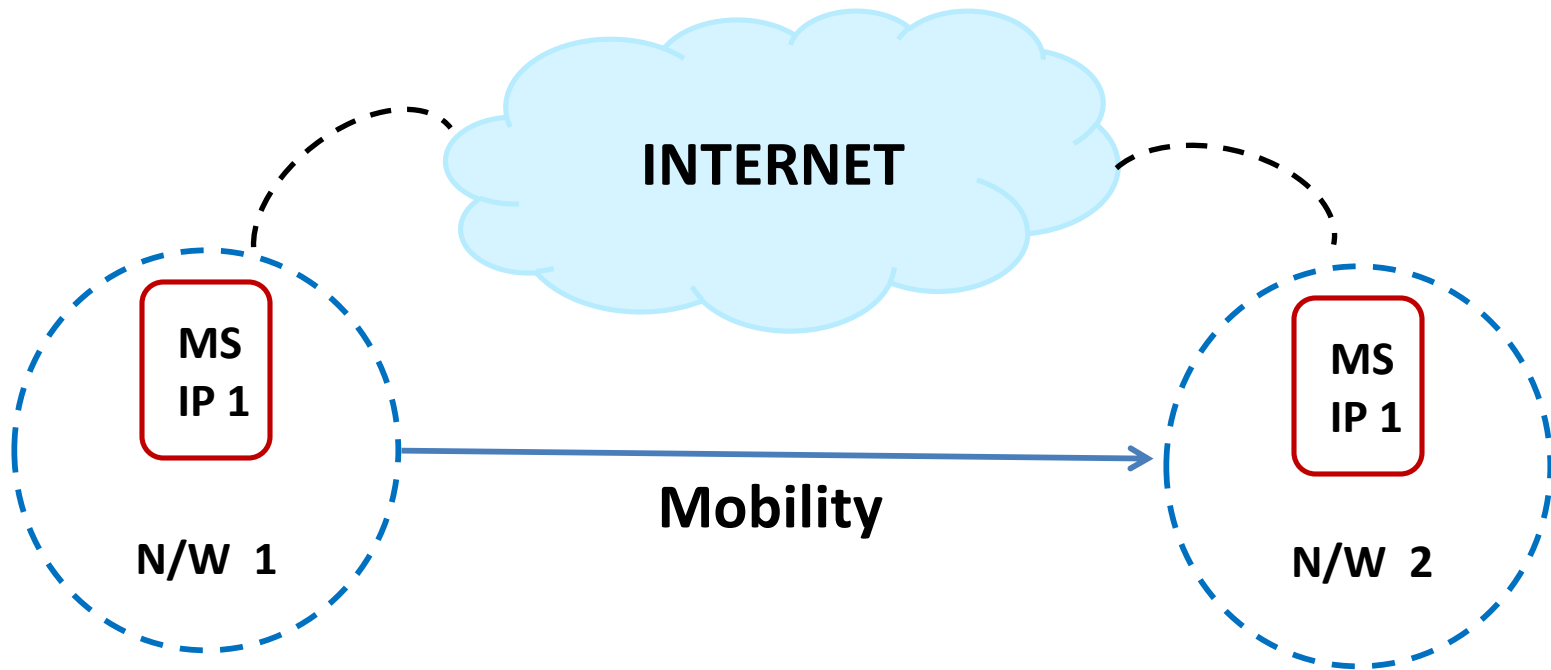
Mobile IP

- **Mobile IP** is a communication protocol (created by extending Internet Protocol, IP) that allows the users to move from one network to another with the same IP address. It allows mobile devices to maintain network connectivity while roaming between different networks (home and foreign) without losing their IP address.
- It is designed by **Internet Engineering Task Force (IETF)**
- It is standard communication protocol that is designed to allow mobile device users to move from one network to another without changing their IP Address.

Mobile IP

- **Why IP is made mobile?**

- IP is made mobile because during mobility IP of device changes and system needs to be reconfigured.



Mobile IP

•Goals of Mobile IP

- Tackles issues with transparency
- Size and frequency of required routing protocols are kept as small as possible as during mobility resources are restricted.
- Simple implementation of mobile node software.

•Requirements

- Compatibility – should not be a change in application or n/w protocol
- Transparency – mobility should be invisible from higher layers.
- Scalability and efficiency – not to be affected by location
- Security – ensure authentication

Mobile IP Terminologies

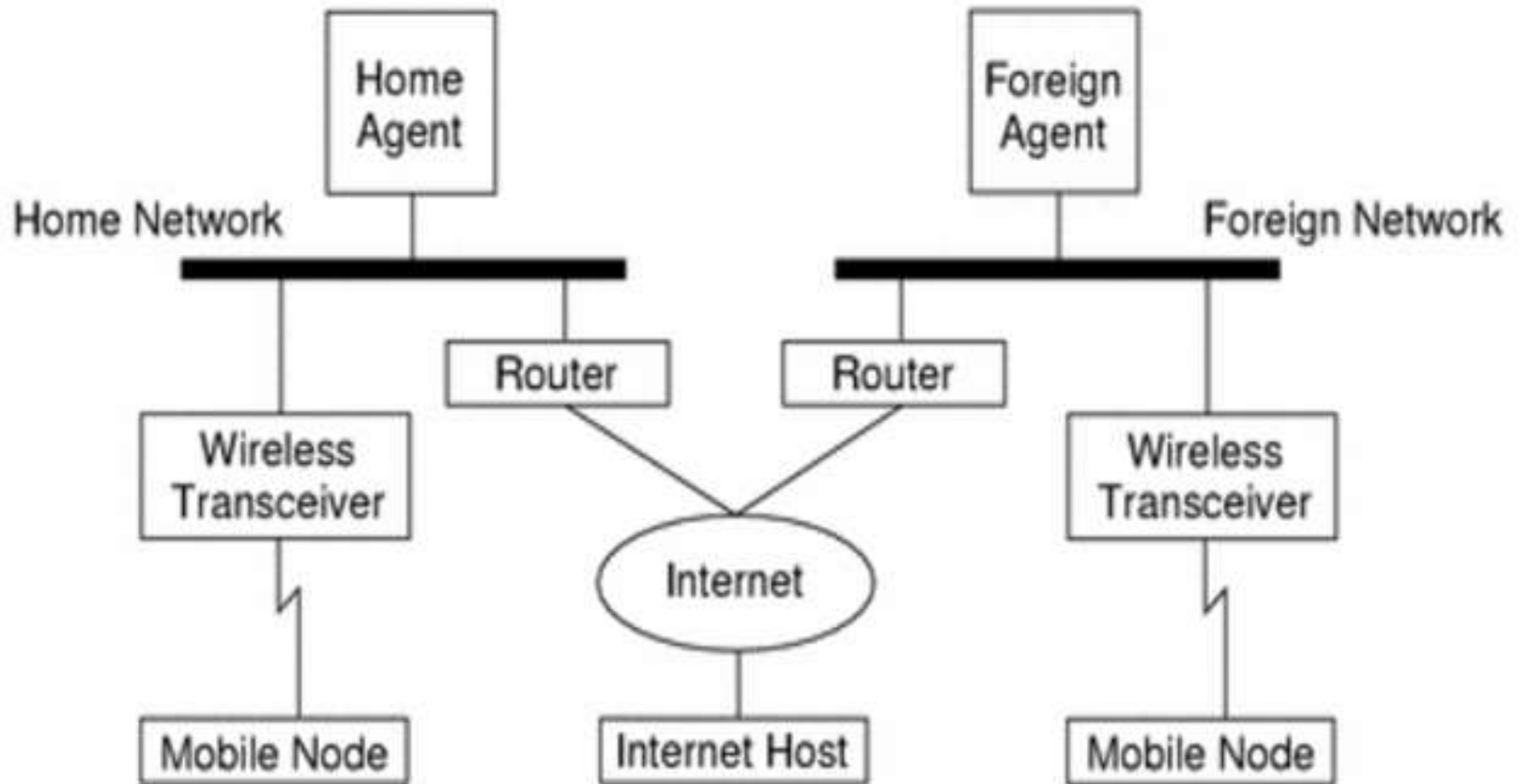


Fig: Mobile IP topology

Mobile IP Terminologies

- **Mobile Node (MN)** is the hand-held communication device that the user carries e.g. Cell phone.
- **Home Network** is a network to which the mobile node originally belongs as per its assigned IP address (home address). No mobile IP support is needed in this network.
- **Home Agent (HA)** is a router in-home network to which the mobile node was originally connected. It encapsulates the data and sends via tunnel when MN is out of HN.
- **Home Address** is the permanent IP address assigned to the mobile node (within its home network).

Mobile IP Terminologies

Foreign Network is the current network to which the mobile node is visiting (away from its home network).

Foreign Agent (FA) is a router in a foreign network to which the mobile node is currently connected. The packets from the home agent are sent to the foreign agent which de-encapsulates and delivers them to the mobile node.

Correspondent Node (CN) is a device on the internet communicating to the mobile node. It can be fixed or mobile.

Mobile IP Terminologies

- **Care-of Address (CoA)** - is the temporary IP address that is allocated by the foreign network, allowing the Home Agent to forward packets to the device even though it is on a different network.

- 1) **Foreign agent CoA** - the FCoA could be located at the FA, i.e., the CoA is an IP address of the FA. The FA is the tunnel end-point so de-encapsulates packets and forwards to the MN.

- 2) **Co-located CoA** - the CoCoA is co-located if the MN temporarily acquired an additional IP address which acts as CoA. This address is now topologically correct, and the tunnel endpoint is at the MN. Co-located addresses can be acquired using services such as DHCP.

Mobile IP Terminologies

- In Mobile IP, a Foreign Agent Care-of Address (FA COA) is the IP address of the foreign agent, while a Co-located Care-of Address (CoCoA) is an IP address assigned to the mobile node itself within the foreign network.

Mobile IP Terminologies

- **Foreign Agent Care-of Address (FA COA):**

- The mobile node uses the foreign agent's IP address as its care-of address.
- All mobile nodes in the foreign network share the same CoA, which is the foreign agent's address.
- The foreign agent acts as a tunnel endpoint and forwards packets to the mobile node.
- The foreign agent also stores information about mobile nodes visiting its network.

Mobile IP Terminologies

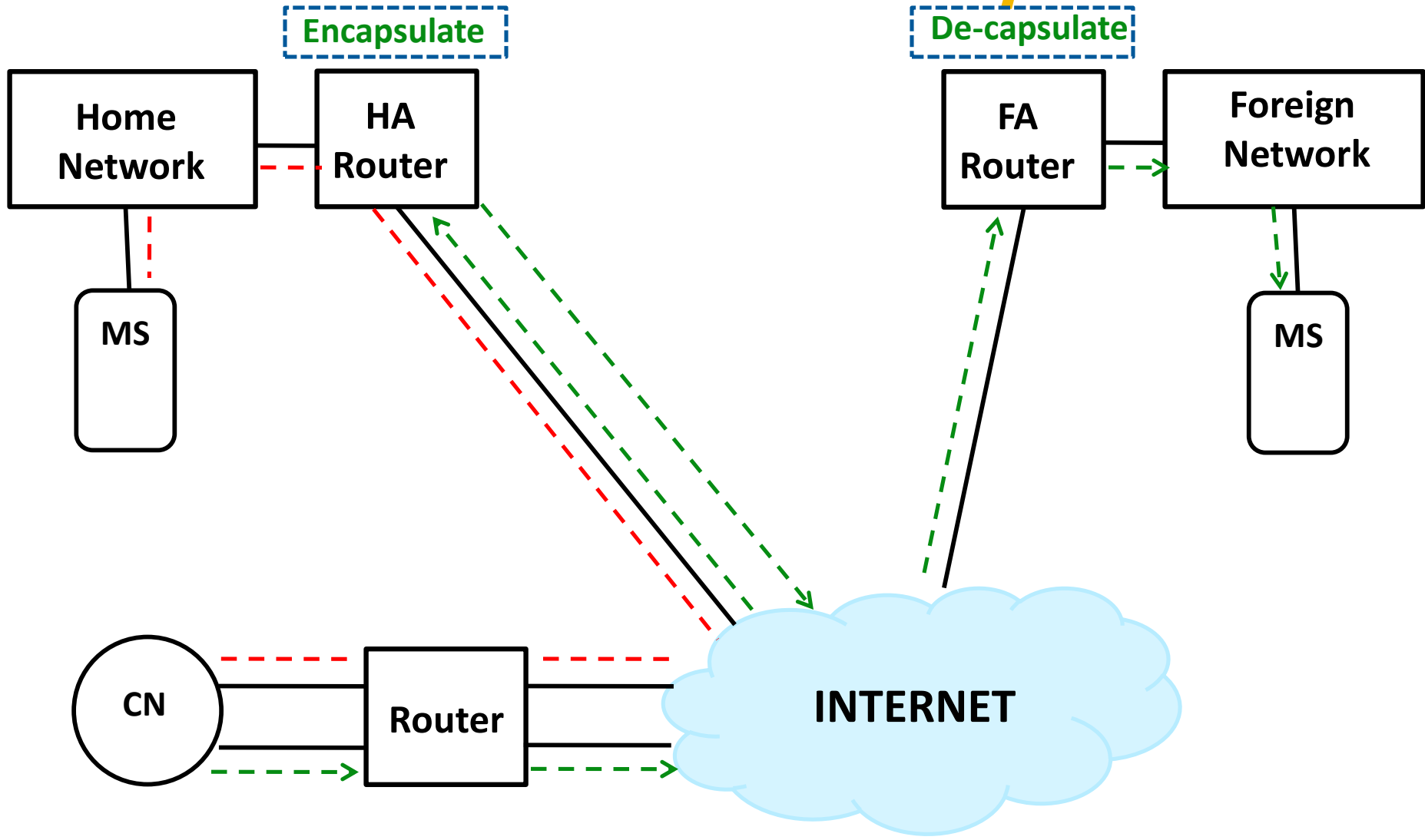
- **Co-located Care-of Address (CoCoA):**

- The mobile node is assigned its own temporary IP address within the foreign network, which acts as its care-of address.
- Each mobile node in the foreign network has its own CoA, usually assigned by a DHCP server.
- This allows the mobile node to function without a foreign agent, for example, in networks that have not yet deployed a foreign agent.
- If a mobile node is using a co-located care-of address, it must be located on the link identified by the network prefix of this care-of address.

Mobile IP Terminologies

- DHCP stands for **Dynamic Host Configuration Protocol**.
- It's a network management protocol that automatically assigns IP addresses to devices on a network.

IP Packet Delivery



Movement of Datagram

- **Step1:** Internet host sends a data gram to the mobile node using the mobile node home address.
- **Step2:**
 - **Case 1 : If MN is in Home Network** – If the MN is present in Home n/w, the datagram is delivered through the normal IP process to the MN.
 - **Case 2 : If MN is in Foreign Network** – If the MN is present in Foreign n/w, the Home Agent forwards the datagram to the foreign Agent after encapsulating the data through a **tunnel**. The Foreign Agent delivers the packet to the MN using FCoA after de-capsulating the datagram.

Key Mechanism in Mobile IP

- Agent Discovery
- Agent Registration
- Tunneling

Key Mechanism in Mobile IP

Agent Discovery

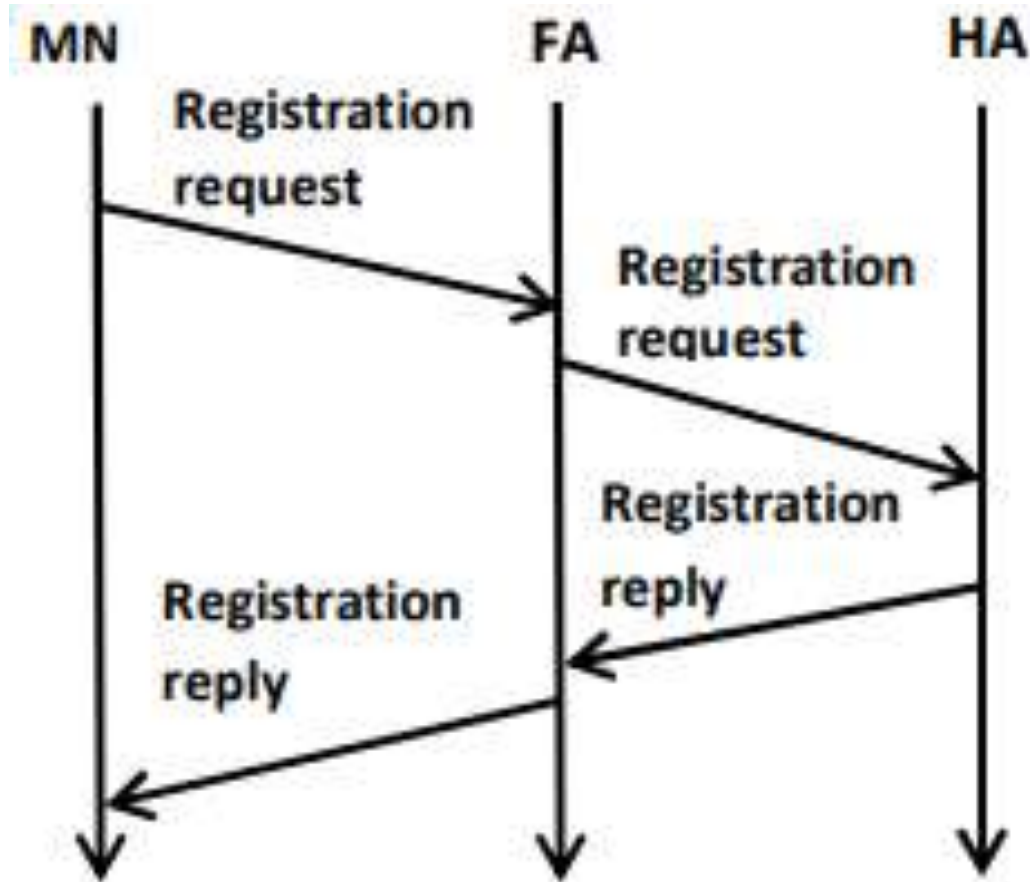
- Agents advertise their presence by periodically broadcasting their agent advertisement messages.
- The mobile node receiving the agent advertisement messages observes whether the message is from its own home agent and determines whether it is in the home network or foreign network.

Key Mechanism in Mobile IP

Agent Registration

- Mobile node after discovering the foreign agent sends a registration request (RREQ) to the foreign agent.
- The foreign agent, in turn, sends the registration request to the home agent with the care-of-address.
- The home agent sends a registration reply (RREP) to the foreign agent. Then it forwards the registration reply to the mobile node and completes the process of registration.

Agent Registration



Key Mechanism in Mobile IP

Tunneling

- It establishes a virtual pipe for the packets available between a tunnel entry and an endpoint.
- It is the process of sending a packet via a tunnel and it is achieved by a mechanism called encapsulation.
- It takes place to forward an IP datagram from the home agent to the care-of-address. Whenever the home agent receives a packet from the correspondent node, it encapsulates the packet with source address as home address and destination as care-of-address.

Tunneling

